

MAT 1013: Discrete Mathematics

Session 9 : Complex Numbers, Polar Form and Roots of Unity

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Topics

- Complex numbers and arithmetic
- Field properties of $\mathbb{C}$
- Complex conjugates and multiplicative inverses
- Geometric interpretation and complex plane
- Polar form and Euler form
- Multiplication as scaling and rotation
- De Moivre's theorem
- Roots of unity and Connection with modular arithmetic
- Solving equations of the form $z^n = w$

Learning Outcomes

By the end of this lesson, students should be able to:

- Define complex numbers and identify real and imaginary parts.
- Perform arithmetic with complex numbers.
- Explain why $\mathbb{C}$ is a field.
- Compute complex conjugates and multiplicative inverses.
- Represent complex numbers geometrically.
- Convert between Cartesian form and polar form.
- Interpret multiplication geometrically as scaling and rotation.
- State and apply De Moivre's theorem.
- Find and visualize roots of unity.
- Explain the connection between roots of unity and arithmetic modulo n .
- Solve equations of the form $z^n = w$.

1. Review of Complex Numbers

The real number system is very powerful, but it does not solve every polynomial equation.

$$x^2 - 1 = 0$$

has real solutions $x = 1$ and $x = -1$.

But

$$x^2 + 1 = 0$$

would require $x^2 = -1$, which has no real solution.

We introduce a new number i , called the imaginary unit, satisfying

$$i^2 = -1.$$

Thus $x^2 + 1 = 0$ has solutions $x = i$ and $x = -i$.

Complex Numbers

Definition: Complex Numbers

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}.$$

If $z = a + bi$, then

$$\operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b.$$

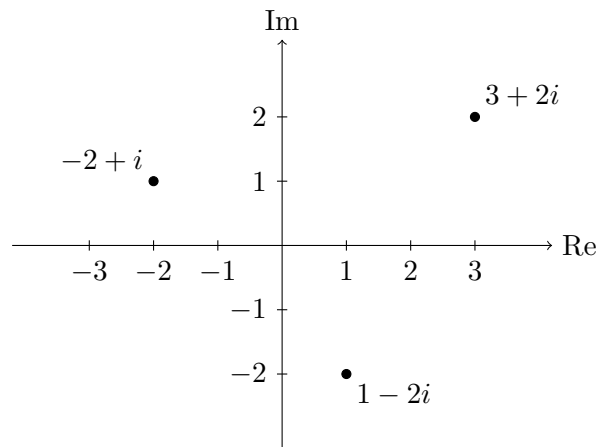
Every real number is complex:

$$a = a + 0i.$$

Thus $\mathbb{R} \subset \mathbb{C}$.

The Complex Plane

A complex number $z = a + bi$ is represented by the point (a, b) . This is the complex plane or Argand plane.



Activity 1:

For each of the following complex numbers, identify its real and imaginary parts and represent it on the complex plane.

1. $3 + 4i$

3. $5 - 6i$

5. $-9i$

2. $-2 + 7i$

4. 8

6. $\cos(\frac{\pi}{3}) + i \sin(\frac{\pi}{3})$

Equality of Complex Numbers

Two complex numbers are equal if and only if they have the same real and imaginary parts. I.e.,

$$a + bi = c + di \iff a = c \text{ and } b = d.$$

Arithmetic of Complex Numbers

Let $z_1 = a + bi$ and $z_2 = c + di$.

$$z_1 + z_2 = (a + c) + (b + d)i.$$

$$z_1 - z_2 = (a - c) + (b - d)i.$$

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Example

$$(2 + 3i)(1 - 2i) = 2 - 4i + 3i - 6i^2 = 8 - i.$$

Activity 2:

Compute:

1. $(3 + 2i) + (4 - 5i)$

3. $(2 + 3i)(1 - 2i)$

5. $(2 - i)^2$

2. $(5 - i) - (2 + 3i)$

4. $(1 + i)^2$

6. $(1 - i)^3$

Complex Conjugate and Multiplicative Inverse

For $z = a + bi$, the complex conjugate is

$$\bar{z} = a - bi.$$

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2.$$

If $z = a + bi \neq 0$, then

$$z^{-1} = \frac{\bar{z}}{z\bar{z}} = \frac{a - bi}{a^2 + b^2}.$$

Example

$$\frac{1}{2 + 3i} = \frac{2 - 3i}{(2 + 3i)(2 - 3i)} = \frac{2 - 3i}{13} = \frac{2}{13} - \frac{3}{13}i.$$

Activity 3:

Find the multiplicative inverse of the following complex numbers.

1. $1 + i$

3. $2 + i$

2. $3 - 4i$

4. $-1 + 2i$

Activity 4:

Verify the following properties for any two complex numbers z and w .

1. $\overline{z + w} = \bar{z} + \bar{w}$

4. $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$

2. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

5. $\bar{z} = z$ if and only if $\operatorname{Im}(z) = 0$

3. $\bar{\bar{z}} = z$

6. $\bar{z} = -z$ if and only if $\operatorname{Re}(z) = 0$

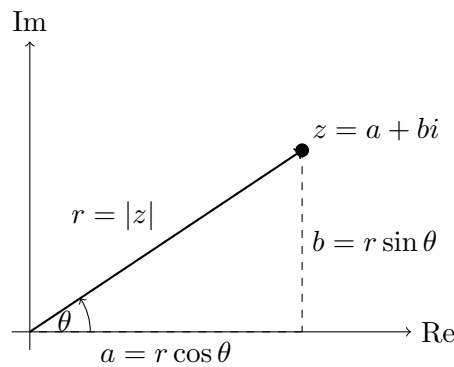
Field Properties of $\mathbb{C}$

For all $z, w, u \in \mathbb{C}$:

1. Closure: $z + w, zw \in \mathbb{C}$.
2. Commutativity: $z + w = w + z, zw = wz$.
3. Associativity: $(z + w) + u = z + (w + u), (zw)u = z(wu)$.
4. Identities: $z + 0 = z, z \cdot 1 = z$.
5. Inverses: $z + (-z) = 0$, and if $z \neq 0$, then $zz^{-1} = 1$.
6. Distributivity: $z(w + u) = zw + zu$.

- The complex numbers form a field.
- Unlike $\mathbb{R}$, the complex numbers cannot be ordered in a way compatible with the field operations. Thus, the complex numbers are not ordered.

Modulus and Argument



For $z = a + bi$, the modulus is

$$|z| = \sqrt{a^2 + b^2}.$$

An argument of z , written $\arg(z)$, is an angle θ from the positive real axis to the vector representing z . If $z = a + bi$, then

$$\tan \theta = \frac{b}{a},$$

with the quadrant taken into account.

Activity 5:

Let $z, w \in \mathbb{C}$.

1. Show that $z\bar{z} = |z|^2$, and hence deduce that $|z| = \sqrt{z\bar{z}}$
2. For $z \neq 0$, verify that $\arg(-z) = \pi + \arg(z)$, and hence deduce that $\tan(\arg(z)) = \tan(\arg(-z))$.
3. Provided $z \neq 0$, what can you say about $\arg(\frac{1}{z})$ and $|\frac{1}{z}|$?
4. Show that $|z \cdot w| = |z| \cdot |w|$.

Polar Representation

If $z = a + bi$, then

$$a = r \cos \theta, \quad b = r \sin \theta,$$

where $r = |z|$. Thus

$$z = r(\cos \theta + i \sin \theta).$$

Example

Convert $z = 1 + i$ to polar form.

$$r = |1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \theta = \frac{\pi}{4}.$$

Thus

$$1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

Euler's Formula and Polar Form

Euler's formula states:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

So the polar form can be written as

$$z = re^{i\theta}.$$

with $r \geq 0$ and $\theta \in [0, 2\pi)$.

Activity 6:

Convert to polar form:

1. $1 + i$

3. $-1 - i$

5. $-3i$

2. $-1 + i$

4. 2

6. $1 - \sqrt{3}i$

Activity 7:

Verify the following:

1. For any angle θ , $|e^{i\theta}| = 1$.

2. If $e^{i\theta} = e^{i\alpha}$, then $\alpha = \theta + 2\pi k$ for some $k \in \mathbb{Z}$.

Multiplication in Polar Form

Let

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2}.$$

Then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

Multiplication of complex numbers has a geometric meaning: moduli multiply and arguments add.

Example

Multiplication by $i = e^{i\pi/2}$ rotates the plane by $\pi/2$ counterclockwise.

2. De Moivre's Theorem

For any integer $n \in \mathbb{Z}$ and $\theta \in \mathbb{R}$,

De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

More generally, if $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$, then Using Euler form:

$$z^n = (re^{i\theta})^n = r^n e^{in\theta}.$$

Thus,

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)).$$

Example

Compute $(1 + i)^8$.

Since

$$1 + i = \sqrt{2}e^{i\pi/4},$$

we get

$$(1 + i)^8 = (\sqrt{2})^8 e^{i8\pi/4} = 16e^{i2\pi} = 16.$$

Activity 8:

Use De Moivre's theorem to compute:

1. $(1 + i)^4$
2. $(\sqrt{3} + i)^3$
3. $(-1 + i)^6$
4. $(2e^{i\pi/6})^5$

3. Roots of Unity

An n -th root of unity is a complex number z satisfying

$$z^n = 1.$$

The n -th roots of unity are

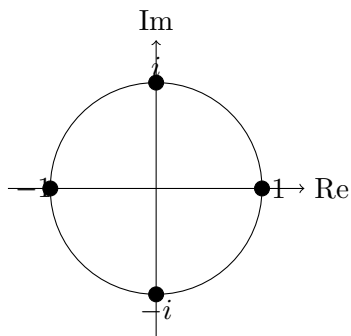
$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

Equivalently,

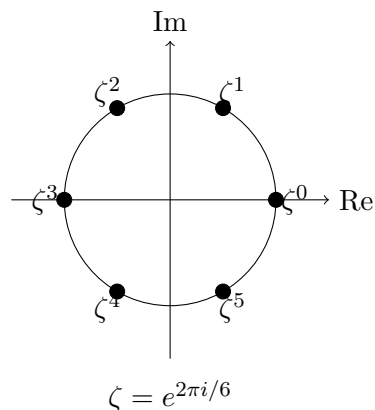
$$z_k = \cos\left(\frac{2\pi k}{n}\right) + i \sin\left(\frac{2\pi k}{n}\right).$$

The n -th roots of unity are equally spaced points on the unit circle. They form the vertices of a regular n -gon.

Example: Fourth Roots of Unity



Example: Sixth Roots of Unity



Activity 9:

Sketch the following roots of unity:

1. $z^3 = 1$

2. $z^5 = 1$

3. $z^8 = 1$

Roots of Unity and Modular Arithmetic

Let

$$\zeta = e^{2\pi i/n}.$$

Then the n -th roots of unity are

$$1, \zeta, \zeta^2, \dots, \zeta^{n-1}.$$

Since $\zeta^n = 1$, we have

$$\zeta^{k+n} = \zeta^k.$$

Thus the exponent behaves modulo n .

$$\zeta^a \zeta^b = \zeta^{a+b}.$$

Since $\zeta^n = 1$, the exponent $a + b$ is taken modulo n :

$$\zeta^a \zeta^b = \zeta^{a+b \pmod n}.$$

The roots of unity form a cyclic structure whose multiplication corresponds to addition modulo n .

Example

Let $\zeta = i$. Then

$$\zeta^0 = 1, \quad \zeta^1 = i, \quad \zeta^2 = -1, \quad \zeta^3 = -i, \quad \zeta^4 = 1.$$

The exponents repeat modulo 4.

Solving Equations of the Form $z^n = w$

We want to solve

$$z^n = w,$$

where w is given.

Write $w = Re^{i\phi}$ with $R \geq 0$.

Suppose $z = re^{i\theta}$ with $r \geq 0$. Then

$$z^n = r^n e^{in\theta}.$$

Thus we need

$$r^n = R$$

and

$$n\theta = \phi + 2\pi k.$$

Because $R \geq 0$ and $r \geq 0$, we have that

$$r = R^{1/n}$$

and

$$\theta = \frac{\phi + 2\pi k}{n}.$$

So the solutions are

$$z_k = R^{1/n} e^{i(\phi + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1.$$

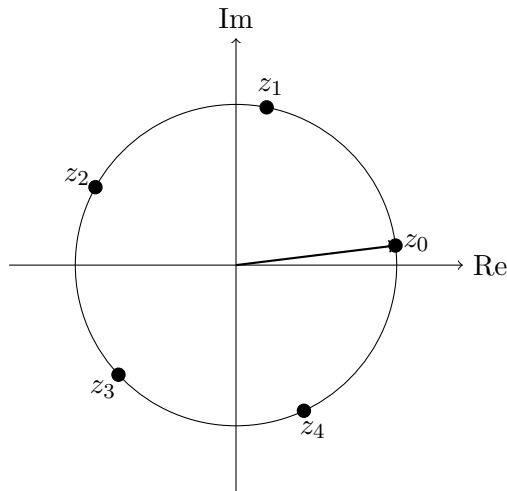
Note that we can write these solutions as

$$z_k = R^{1/n} e^{i\phi} e^{i2\pi k/n} = z_0 \zeta_k,$$

where $\zeta_k = e^{i2\pi k/n}$ is an n -th root of unity and $z_0 = R^{1/n} e^{i\phi}$ is a “*particular solution*” to this equation.

The solutions of $z^n = w$ are equally spaced points on a circle of radius $R^{1/n}$. They form the vertices of a regular n -gon.

Example: Solutions of $z^5 = w$



Five solutions of $z^5 = w$, equally spaced on a circle

Example: Solve $z^3 = 8$

Write

$$8 = 8e^{i0}.$$

Here $R = 8$ and $\phi = 0$.

The cube roots have modulus

$$R^{1/3} = 2.$$

The arguments are

$$\theta_k = \frac{0 + 2\pi k}{3} = \frac{2\pi k}{3}, \quad k = 0, 1, 2.$$

Therefore,

$$\begin{aligned} z_0 &= 2, \\ z_1 &= 2e^{2\pi i/3} = -1 + \sqrt{3}i, \\ z_2 &= 2e^{4\pi i/3} = -1 - \sqrt{3}i. \end{aligned}$$

Example: Solve $z^4 = -16$

Write

$$-16 = 16e^{i\pi}.$$

Then $R = 16$ and $\phi = \pi$. The fourth roots have modulus

$$16^{1/4} = 2.$$

The arguments are

$$\theta_k = \frac{\pi + 2\pi k}{4}, \quad k = 0, 1, 2, 3.$$

Thus the arguments are

$$\frac{\pi}{4}, \quad \frac{3\pi}{4}, \quad \frac{5\pi}{4}, \quad \frac{7\pi}{4}.$$

The solutions are

$$2e^{i\pi/4}, \quad 2e^{3\pi i/4}, \quad 2e^{5\pi i/4}, \quad 2e^{7\pi i/4}.$$

In rectangular form:

$$\sqrt{2} + \sqrt{2}i, \quad -\sqrt{2} + \sqrt{2}i, \quad -\sqrt{2} - \sqrt{2}i, \quad \sqrt{2} - \sqrt{2}i.$$

4. Exercises

1. Arithmetic: Evaluate the following and write your answer in $a + bi$ form.

$$\begin{array}{ll} \text{(a)} (3 + 2i)(4 - i) & \text{(c)} \frac{1}{1 + i} \\ \text{(b)} (1 + i)^3 & \text{(d)} \frac{3}{2 - i} \end{array}$$

2. Polar Form: Convert the following complex numbers to polar form.

$$\begin{array}{lll} \text{(a)} 1 + i & \text{(c)} \sqrt{3} + i & \text{(e)} 2 - 2i \\ \text{(b)} -1 + i & \text{(d)} -2 & \text{(f)} -i \end{array}$$

3. De Moivre's Theorem: Use De Moivre's theorem to compute the following.

$$\begin{array}{lll} \text{(a)} (1 + i)^8 & \text{(b)} (\sqrt{3} + i)^6 & \text{(c)} \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right)^5 \end{array}$$

4. Roots of Unity: Find and sketch the following.

- (a) the third roots of unity,
- (b) the fourth roots of unity,
- (c) the sixth roots of unity.

5. Solving $z^n = w$: Solve the following equations.

$$\begin{array}{lll} \text{(a)} z^3 = 1 & \text{(c)} z^4 = 16i & \text{(e)} z^5 = 32 \\ \text{(b)} z^3 = 8 & \text{(d)} z^4 = -16 & \text{(f)} z^3 = i \end{array}$$

6. Proof Exercises: Prove the following statements for $z, w \in \mathbb{C}$.

- (a) Prove that $|zw| = |z||w|$.
- (b) Prove that $\overline{z + w} = \bar{z} + \bar{w}$.
- (c) Prove that $\overline{z\bar{w}} = \bar{z}w$.
- (d) Prove that if $z \neq 0$, then $z^{-1} = \frac{\bar{z}}{|z|^2}$.
- (e) Prove De Moivre's theorem for positive integers n using induction.
- (f) Prove that the sum of all n -th roots of unity is zero for $n > 1$.

7. Some challenges:

- (a) Show that multiplication by $e^{i\theta}$ rotates every complex number by angle θ .
- (b) Let $\zeta = e^{2\pi i/n}$. Prove that

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1} = 0.$$

- (c) Solve $z^6 = -64$.
- (d) Solve $z^5 = 1 + i$.
- (e) Show that the product of all n -th roots of unity is $(-1)^{n-1}$.
- (f) Find all solutions of $z^4 + 16 = 0$.

- (g) Determine all solutions to $z^2 = i$. (These are the possible values for $\sqrt{i}$)
8. Check Learning Outcomes:
- (a) Convert $1 + i$ to polar form.
 - (b) Compute $(1 + i)^4$ using De Moivre's theorem.
 - (c) List the fourth roots of unity.
 - (d) Solve $z^3 = 8$.
9. Investigate:
- (a) Euler's formula and Euler's identity.
 - (b) Roots of unity and regular polygons.
 - (c) Explain why roots of unity are connected to arithmetic modulo n .
 - (d) Complex numbers in electrical engineering.
 - (e) Roots of unity and the Fast Fourier Transform.
 - (f) Cyclic groups and roots of unity.